

Chapter 8 Homework: Heteroskedasticity

Claire

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Exercise 8.6

Consider the wage equation:

$$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + e_i$$

(XR8.6a)

where wage is measured in dollars per hour, education and experience in years, and METRO = 1 if the person lives in a metropolitan area.

```
load(url("http://www.principlesofeconometrics.com/poe5/data/rdata/cps5_small.rdata"))
head(cps5_small)
```

	black	educ	exper	faminc	female	metro	midwest	south	wage
	<int>	<int>	<int>	<int>	<int>	<int>	<int>	<int>	<dbl>
1	0	13	45	0	1	1	0	0	44.44
2	0	14	25	45351	1	1	1	0	16.00
3	0	18	27	91946	1	1	0	0	15.38
4	0	13	42	48370	0	1	1	0	13.54
5	0	13	41	10000	1	1	0	0	25.00
6	0	16	26	151308	1	1	0	0	24.05

6 rows | 1-10 of 11 columns

```
str(cps5_small)
```

```
## 'data.frame': 1200 obs. of 10 variables:
## $ black : int 0 0 0 0 0 0 0 0 0 0 ...
## $ educ : int 13 14 18 13 13 16 16 18 21 14 ...
## $ exper : int 45 25 27 42 41 26 11 15 32 12 ...
## $ faminc : int 0 45351 91946 48370 10000 151308 110461 0 67084 14000 ...
## $ female : int 1 1 1 0 1 1 1 1 0 0 ...
## $ metro : int 1 1 1 1 1 1 1 1 1 0 ...
## $ midwest: int 0 1 0 1 0 0 0 1 1 0 ...
## $ south : int 0 0 0 0 0 0 1 0 0 0 ...
## $ wage : num 44.4 16 15.4 13.5 25 ...
## $ west : int 1 0 0 0 1 0 0 0 0 1 ...
## - attr(*, "var.labels")= chr [1:10] "=1 if black" "years of education" "potential experience" "other family income, $" ...
```

Part (a): OLS Estimation

```
model_ols <- lm(wage ~ educ + exper + metro, data = cps5_small)
summary(model_ols)
```

```
##
## Call:
## lm(formula = wage ~ educ + exper + metro, data = cps5_small)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -29.383  -7.930  -2.882   5.553  189.262
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -20.22038    2.31055  -8.751  < 2e-16 ***
## educ         2.54772    0.13548  18.805  < 2e-16 ***
## exper        0.20443    0.02946   6.940 6.42e-12 ***
## metro        3.52722    1.00305   3.516 0.000454 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.24 on 1196 degrees of freedom
## Multiple R-squared:  0.2444, Adjusted R-squared:  0.2425
## F-statistic: 129 on 3 and 1196 DF, p-value: < 2.2e-16
```

Interpretation: - Each additional year of education increases expected wage by approximately $\hat{\beta}_2$. - Each additional year of experience increases wage by approximately $\hat{\beta}_3$. - Living in a metro area adds approximately $\hat{\beta}_4$ per hour.

Part (b): Add FEMALE indicator; check for heteroskedasticity

```
# Extended model with FEMALE
model_ext <- lm(wage ~ educ + exper + metro + female, data = cps5_small)
summary(model_ext)
```

```
##
## Call:
## lm(formula = wage ~ educ + exper + metro + female, data = cps5_small)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -31.684  -7.822  -2.528   5.115  187.134
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -19.71781    2.28209  -8.640  < 2e-16 ***
## educ         2.63581    0.13460  19.583  < 2e-16 ***
## exper        0.20423    0.02907   7.025 3.59e-12 ***
## metro        3.74974    0.99073   3.785 0.000161 ***
## female      -4.39075    0.76629  -5.730 1.27e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 13.07 on 1195 degrees of freedom
## Multiple R-squared:  0.2646, Adjusted R-squared:  0.2621
## F-statistic: 107.5 on 4 and 1195 DF,  p-value: < 2.2e-16
```

```
# Save residuals for heteroskedasticity test
ehat <- residuals(model_ext)
ehat2 <- ehat^2

# Breusch-Pagan (NR^2) test: regress ehat^2 on regressors
aux_model <- lm(ehat2 ~ educ + exper + metro + female, data = cps5_small)
N <- nrow(cps5_small)
R2 <- summary(aux_model)$r.squared
LM <- N * R2

cat("\nBreusch-Pagan LM test statistic:", round(LM, 4), "\n")
```

```
##
## Breusch-Pagan LM test statistic: 16.4026
```

```
cat("Chi^2 critical value (5%, df=4):", round(qchisq(0.95, 4), 4), "\n")
```

```
## Chi^2 critical value (5%, df=4): 9.4877
```

```
cat("Reject H0 (homoskedasticity)?", LM > qchisq(0.95, 4), "\n")
```

```
## Reject H0 (homoskedasticity)? TRUE
```

Part (c): White test for heteroskedasticity

```
library(lmtest)
library(sandwich)

# White test (using squares and cross-products of regressors)
white_test <- bptest(model_ext, ~ educ*exper + educ*metro + educ*female +
                      exper*metro + exper*female + metro*female +
                      I(educ^2) + I(exper^2), data = cps5_small)

print(white_test)
```

```
##
## studentized Breusch-Pagan test
##
## data: model_ext
## BP = 21.815, df = 12, p-value = 0.03965
```

```
cat("Reject H0 (homoskedasticity) at 5%?", white_test$p.value < 0.05, "\n")
```

```
## Reject H0 (homoskedasticity) at 5%? TRUE
```

Part (d): OLS with White heteroskedasticity-robust standard errors

```
coeftest(model_ext, vcov = vcovHC(model_ext, type = "HC1"))
```

```
##
## t test of coefficients:
##
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -19.717811   2.413394 -8.1702 7.774e-16 ***
## educ         2.635805    0.150282 17.5390 < 2.2e-16 ***
## exper        0.204231    0.031015  6.5849 6.802e-11 ***
## metro        3.749740    0.780850  4.8021 1.769e-06 ***
## female      -4.390747    0.764614 -5.7424 1.183e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
# 95% CI comparison
ci_ols    <- confint(model_ext)
ci_robust <- coefci(model_ext, vcov = vcovHC(model_ext, type = "HC1"))

cat("\nComparison of interval widths (OLS vs Robust):\n")
```

```
##
## Comparison of interval widths (OLS vs Robust):
```

```
width_ols    <- ci_ols[, 2] - ci_ols[, 1]
width_robust <- ci_robust[, 2] - ci_robust[, 1]
comparison   <- data.frame(OLS_width = round(width_ols, 4),
                           Robust_width = round(width_robust, 4),
                           Narrower = width_robust < width_ols)

print(comparison)
```

```
##           OLS_width Robust_width Narrower
## (Intercept)   8.9547      9.4699   FALSE
## educ         0.5281      0.5897   FALSE
## exper        0.1141      0.1217   FALSE
## metro        3.8875      3.0640    TRUE
## female       3.0068      3.0003    TRUE
```

Part (e): GLS — dividing by SQFT (assuming $\sigma_i^2 = \sigma^2 \cdot SQFT_i$)

Note: The textbook suggests GLS by assuming heteroskedasticity proportional to a variable. Here we implement FGLS using the auxiliary regression approach.

```
# FGLS: estimate variance function
ln_ghat2 <- log(ghat2)
aux_fgls <- lm(ln_ghat2 ~ educ + exper + metro + female, data = cps5_small)
sigma_hat <- exp(fitted(aux_fgls))

# GLS weights
weights_gls <- 1 / sigma_hat

model_fgls <- lm(wage ~ educ + exper + metro + female,
                 data = cps5_small, weights = weights_gls)
summary(model_fgls)
```

```
##
## Call:
## lm(formula = wage ~ educ + exper + metro + female, data = cps5_small,
##     weights = weights_gls)
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -3.6684 -1.3718 -0.4555  0.9060 21.5165
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -9.55524    1.54324  -6.192 8.17e-10 ***
## educ         1.91573    0.09487  20.192 < 2e-16 ***
## exper        0.20648    0.02359   8.755 < 2e-16 ***
## metro        2.69392    0.68510   3.932 8.90e-05 ***
## female      -3.82387    0.62186  -6.149 1.06e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.011 on 1195 degrees of freedom
## Multiple R-squared:  0.2671, Adjusted R-squared:  0.2646
## F-statistic: 108.9 on 4 and 1195 DF,  p-value: < 2.2e-16
```

Part (f): Compare OLS, robust OLS, and FGLS estimates

```
library(stargazer)
stargazer(model_ext, model_fgls,
           se = list(sqrt(diag(vcovHC(model_ext, type="HC1"))), NULL),
           column.labels = c("OLS (Robust SE)", "FGLS"),
           type = "text", title = "Wage Equation Estimates")
```

```
##
## Wage Equation Estimates
## =====
##                               Dependent variable:
##                               -----
##                               wage
##                               OLS (Robust SE)      FGLS
##                               (1)                (2)
## -----
## educ                        2.636***           1.916***
##                               (0.150)           (0.095)
##
## exper                       0.204***           0.206***
##                               (0.031)           (0.024)
##
## metro                       3.750***           2.694***
##                               (0.781)           (0.685)
##
## female                     -4.391***           -3.824***
##                               (0.765)           (0.622)
##
## Constant                   -19.718***           -9.555***
##                               (2.413)           (1.543)
## -----
## Observations                1,200             1,200
## R2                          0.265             0.267
## Adjusted R2                 0.262             0.265
## Residual Std. Error (df = 1195) 13.071         2.011
## F Statistic (df = 4; 1195)    107.495***      108.877***
## =====
## Note:                        *p<0.1; **p<0.05; ***p<0.01
```

Exercise 8.16

A sample of 200 Chicago households was taken to investigate how far American households tend to travel when they take a vacation:

$$MILES = \beta_1 + \beta_2 INCOME + \beta_3 AGE + \beta_4 KIDS + e$$

```
load(url("http://www.principlesofeconometrics.com/poe5/data/rdata/vacation.rdata"))
head(vacation)
```

	miles <int>	income <int>	age <int>	kids <int>
1	902	41	26	0
2	491	31	38	3
3	1841	87	40	2
4	406	54	48	4

	miles <int>	income <int>	age <int>	kids <int>
5	0	77	43	4
6	1899	70	55	2

6 rows

```
str(vacation)
```

```
## 'data.frame': 200 obs. of 4 variables:
## $ miles : int 902 491 1841 406 0 1899 724 1404 1367 560 ...
## $ income: int 41 31 87 54 77 70 43 40 69 38 ...
## $ age : int 26 38 40 48 43 55 24 39 55 29 ...
## $ kids : int 0 3 2 4 4 2 0 0 2 2 ...
## - attr(*, "var.labels")= chr [1:4] "miles traveled per year" "annual income ($100
0)" "average age of adult members of household" "number of children in household"
```

Part (a): OLS estimation and 95% CI for KIDS effect

```
model_vac <- lm(miles ~ income + age + kids, data = vacation)
summary(model_vac)
```

```
##
## Call:
## lm(formula = miles ~ income + age + kids, data = vacation)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1198.14  -295.31   17.98   287.54  1549.41
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -391.548    169.775  -2.306   0.0221 *
## income         14.201     1.800    7.889 2.10e-13 ***
## age           15.741     3.757    4.189 4.23e-05 ***
## kids          -81.826    27.130   -3.016  0.0029 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 452.3 on 196 degrees of freedom
## Multiple R-squared:  0.3406, Adjusted R-squared:  0.3305
## F-statistic: 33.75 on 3 and 196 DF, p-value: < 2.2e-16
```

```
ci <- confint(model_vac, level = 0.95)
cat("\n95% CI for KIDS coefficient:\n")
```

```
##
## 95% CI for KIDS coefficient:
```



```
print(ci["kids", ])
```

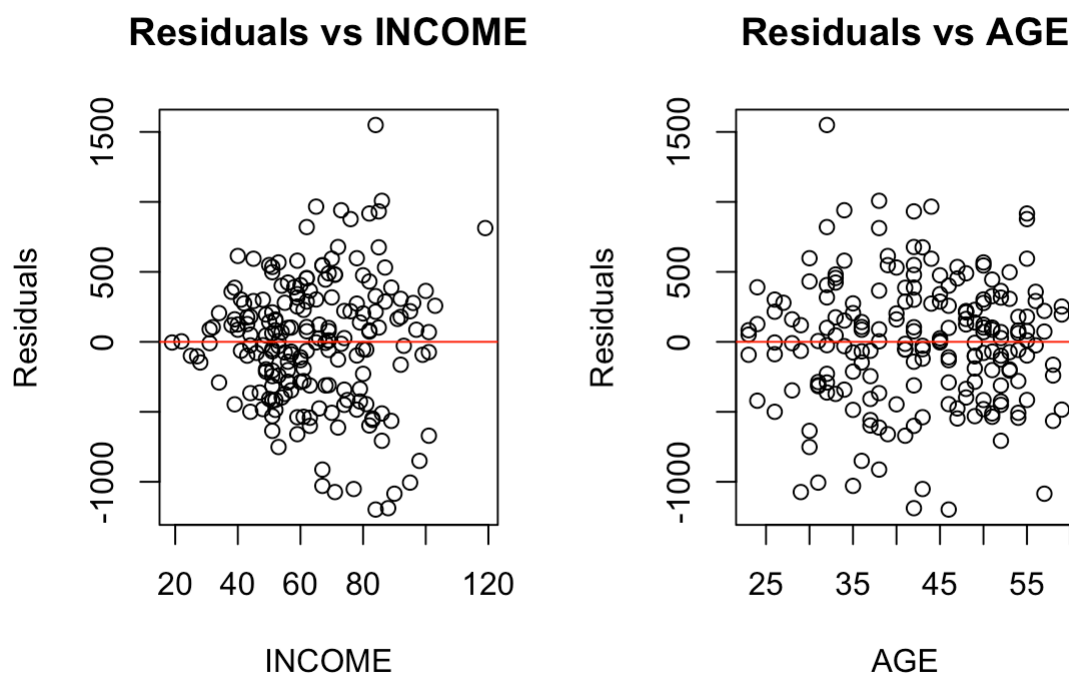
```
##          2.5 %          97.5 %
## -135.32981  -28.32302
```

Interpretation: Holding INCOME and AGE constant, each additional child is associated with a change of -81.83 miles driven per year.

Part (b): Residual plots vs INCOME and AGE

```
par(mfrow = c(1, 2))
plot(vacation$income, residuals(model_vac),
     xlab = "INCOME", ylab = "Residuals", main = "Residuals vs INCOME")
abline(h = 0, col = "red")

plot(vacation$age, residuals(model_vac),
     xlab = "AGE", ylab = "Residuals", main = "Residuals vs AGE")
abline(h = 0, col = "red")
```



```
par(mfrow = c(1, 1))
```

Observation: If the spread of residuals increases with INCOME, this indicates heteroskedasticity.

Part (c): Goldfeld–Quandt test (sorted by

INCOME)

```
# Sort by income
vac_sorted <- vacation[order(vacation$income), ]
N <- nrow(vac_sorted)
n90 <- 90

# Lower 90 observations
lower90 <- vac_sorted[1:n90, ]
upper90 <- vac_sorted[(N - n90 + 1):N, ]

model_lower <- lm(miles ~ income + age + kids, data = lower90)
model_upper <- lm(miles ~ income + age + kids, data = upper90)

SSE_lower <- sum(residuals(model_lower)^2)
SSE_upper <- sum(residuals(model_upper)^2)

df1 <- n90 - 4; df2 <- n90 - 4
GQ <- (SSE_upper / df1) / (SSE_lower / df2)

p_val <- pf(GQ, df1 = df1, df2 = df2, lower.tail = FALSE)

cat("Goldfeld-Quandt F statistic:", round(GQ, 4), "\n")
```

```
## Goldfeld-Quandt F statistic: 3.1041
```

```
cat("p-value:", round(p_val, 4), "\n")
```

```
## p-value: 0
```

```
cat("5% critical value:", round(qf(0.95, df1, df2), 4), "\n")
```

```
## 5% critical value: 1.4286
```

```
cat("Reject H0?", GQ > qf(0.95, df1, df2), "\n")
```

```
## Reject H0? TRUE
```

Part (d): OLS with heteroskedasticity-robust standard errors

```
robust_se <- coeftest(model_vac, vcov = vcovHC(model_vac, type = "HC1"))
print(robust_se)
```

```
##
## t test of coefficients:
##
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) -391.5480   142.6548  -2.7447 0.0066190 **
## income      14.2013     1.9389   7.3246 6.083e-12 ***
## age         15.7409     3.9657   3.9692 0.0001011 ***
## kids        -81.8264    29.1544  -2.8067 0.0055112 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
ci_robust <- coefci(model_vac, vcov = vcovHC(model_vac, type = "HC1"))
cat("\n95% CI for KIDS (robust):\n")
```

```
##
## 95% CI for KIDS (robust):
```

```
print(ci_robust["kids", ])
```

```
##           2.5 %           97.5 %
## -139.32297    -24.32986
```

Part (e): GLS assuming $\sigma_i^2 = \sigma^2 \cdot INCOME_i^2$

Under this assumption, divide all variables by $INCOME_i$:

```
# GLS with variance proportional to INCOME^2
model_gls <- lm(miles ~ income + age + kids, data = vacation,
               weights = 1 / vacation$income^2)
summary(model_gls)
```

```
##
## Call:
## lm(formula = miles ~ income + age + kids, data = vacation, weights = 1/vacation$income^2)
##
## Weighted Residuals:
##      Min       1Q   Median       3Q      Max
## -15.1907  -4.9555   0.2488   4.3832  18.5462
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -424.996    121.444  -3.500 0.000577 ***
## income         13.947     1.481   9.420 < 2e-16 ***
## age           16.717     3.025   5.527 1.03e-07 ***
## kids          -76.806    21.848  -3.515 0.000545 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 6.765 on 196 degrees of freedom
## Multiple R-squared:  0.4573, Adjusted R-squared:  0.449
## F-statistic: 55.06 on 3 and 196 DF,  p-value: < 2.2e-16
```

```
# CI for KIDS (conventional GLS SE)
ci_gls <- confint(model_gls, level = 0.95)
cat("\n95% CI for KIDS (GLS, conventional):\n")
```

```
##
## 95% CI for KIDS (GLS, conventional):
```

```
print(ci_gls["kids", ])
```

```
##      2.5 %      97.5 %
## -119.89450  -33.71808
```

```
# CI for KIDS (robust GLS SE)
ci_gls_rob <- coefci(model_gls, vcov = vcovHC(model_gls, type = "HC1"))
cat("\n95% CI for KIDS (GLS, robust):\n")
```

```
##
## 95% CI for KIDS (GLS, robust):
```

```
print(ci_gls_rob["kids", ])
```

```
##      2.5 %      97.5 %
## -121.41339  -32.19919
```

```
# Comparison of all interval estimates
cat("\n--- Summary Comparison for KIDS Coefficient ---\n")
```

```
##  
## --- Summary Comparison for KIDS Coefficient ---
```

```
cat("OLS:           [", round(ci["kids",1],2), ",", round(ci["kids",2],2), "]\n")
```

```
## OLS:           [ -135.33 , -28.32 ]
```

```
cat("OLS Robust:    [", round(ci_robust["kids",1],2), ",", round(ci_robust["kids",2],  
2), "]\n")
```

```
## OLS Robust:    [ -139.32 , -24.33 ]
```

```
cat("GLS:           [", round(ci_gls["kids",1],2), ",", round(ci_gls["kids",2],2), "]\n")
```

```
## GLS:           [ -119.89 , -33.72 ]
```

```
cat("GLS Robust:    [", round(ci_gls_rob["kids",1],2), ",", round(ci_gls_rob["kids",  
2],2), "]\n")
```

```
## GLS Robust:    [ -121.41 , -32.2 ]
```

Interpretation: The GLS estimator assumes $\sigma_i^2 = \sigma^2 INCOME_i^2$. If this form is correct, GLS is more efficient than OLS, yielding narrower confidence intervals.